

Practice Quiz 01B — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 01

Tautologies, contradictions, and contingent formulas.

1. Make one complete four-row truth table for

$$R = (P \vee Q) \vee \neg P \quad \text{and} \quad S = (P \vee Q) \wedge \neg P.$$

Use the table to classify each formula as a tautology, a contradiction, or neither, and explain how its column supports your classification.

Solution

P	Q	$P \vee Q$	$\neg P$	R	S
F	F	F	T	T	F
F	T	T	T	T	T
T	F	T	F	T	F
T	T	T	F	T	F

The formula R is a tautology because its column is true in every row. The formula S is neither a tautology nor a contradiction because its column contains both truth values: it is true in the second row and false in the other three. ■